Recovering signs from a binomial digit

An explicit quartic consistency equation in 247 integer variables

Research prepared with AI assistance
For Vladimir Reshetnikov and the ProveIt repository

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Abstract

We give a completely specified ordinary polynomial $P \in \mathbb{Z}[z_1, \ldots, z_{247}]$ of degree four, with 4,755 collected monomials and coefficient height 8,815, such that ZFC proves that P has no integer zero if and only if ZFC is consistent. The polynomial is the sum of squares of 182 explicitly printed quadratic residuals. Its mathematical reduction is proved by composition with the retained, pinned C222 refutation calibration.

The new step recovers the sign of the factorial binomial digit before using the usual ceiling-sign certificates. A retained denominator-ratio equation forces an early exponent bound, and the self-sized radix then separates the positive binomial digit from all signed alternatives. Three scalar nonnegativity certificates can consequently be replaced by one product certificate and a cofinally available square restriction on the radix. Combined with an inherited deterministic Pell comparison center, this removes five variables from the 252-integer common-remainder predecessor. Certified integral changes of coordinates and residual basis give the final height and support.

This is a measured improvement of a particular small-coefficient direct quartic, not a variable-count record or a proof of global minimality. The new unbounded argument is an ordinary mathematical proof; a fresh end-to-end proof-assistant formalization is not claimed. Independent finite checks validate the exact source changes, integral restorations, all coefficients, and every residual printed here.

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1 The delivered equation and the meaning of simplicity

The deliverable is the single equation

$$P(z_1,\dots,z_{247}) := \sum_{j=1}^{182} R_j(z_1,\dots,z_{247})^2 = 0, \qquad (z_1,\dots,z_{247}) \in \mathbb{Z}^{247}. \tag{1}$$

Appendix A prints every R_j. The file `equations/quartic247/equation.json` gives the same finite list, and `equations/quartic247/expanded_polynomial.json` gives every collected coefficient of P. All variables are unrestricted integers. There are no suppressed parameters, variable-exponent operations, factorial operations, proof predicates, or quantified helper functions in (1). The notation for such objects below is used only to prove the meaning of ordinary arithmetic registers.

Theorem 1. *Let P be the literal polynomial (1). With the syntactic axiomatization of ZFC used by the retained ProveIt calibration, ZFC proves*

$$\big(\exists z_1,\dots,z_{247}\in\mathbb{Z},\ P(z_1,\dots,z_{247})=0\big) \iff \textit{there is a finite ZFC refutation.}$$

Equivalently,

$$\mathsf{ZFC} \vdash \Big[\big(\forall \mathbf{z}\in\mathbb{Z}^{247},\ P(\mathbf{z})\neq 0\big) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC})\Big].$$

Its degree is exactly 4*, its support has exactly* 4 755 *monomials, and its maximum absolute coefficient is* 8 815*.*

Here consistency means the absence of a finite syntactic derivation of contradiction in the fixed first-order ZFC theory. We do not assert either side of the equivalence separately. The theorem does not conflict with the second incompleteness theorem: if ZFC is consistent, the construction does not supply a ZFC proof of the universal nonvanishing statement.

Endpoint over unrestricted integers	Variables	Rows	Degree	Terms	Height
Common-remainder predecessor	252	182	4	5,340	128,446
New source after integral projection	247	182	4	5,310	128,446
Delivered coordinate and residual basis	247	182	4	4,755	8,815
Prior shared-residue comparison	260	186	4	4,770	9,191

The comparison is pinned to repository commit `3ecd6a6cf53d47519228328567f45fb7ee9781d9`, including the incoming archives listed in `SOURCE_LEDGER.json`. Relative to the matched 252-variable predecessor, the delivered polynomial removes five variables and 585 collected monomials, and lowers the height by 119,631. Relative to the prior 260-variable small-height endpoint, it also improves all three displayed size measures.

There is no canonical total ordering of Diophantine equations by simplicity. The repository already contains a quartic in 37 unrestricted integers with large fixed coefficients, as well as still smaller variable counts at larger degrees [1]. Those are not displaced as variable-count achievements by this result. Our concrete objective is a direct, fully expanded, modest-coefficient quartic with transparent arithmetic witnesses. The route uses the numerical calculus C222 directly; it does not encode a K215 computation or instantiate Jones's universal polynomial. C222's retained logical calibration is nevertheless an essential inherited dependency.

For small integrity checks on the literal expansion, the constant coefficient is 3494, the coefficient of z_1^4 is 13, and the coefficient of $z_{12}^2 z_{14}^2$ is 8815. These are not a substitute for the complete coefficient file.

2 The retained arithmetic interface

We begin with the direct common-remainder construction [2]. Its semantic registers satisfy the following ordinary identities before their intended signs have been established:

$$t = 16 + h_0^2, \qquad B = r + s + 1, \tag{2}$$

$$BF = V + s, \qquad y = V(V-1), \tag{3}$$

$$q = B + o + 1, \qquad x = 1 + [1 + (t+1)y]u_t, \tag{4}$$

$$B\mu = F^2 + C_0 + 1, \qquad H_0 = 1 + \sum_{i=1}^{9} a_i. \tag{5}$$

The a_i are the nine actual encoded representatives, including C_0. They are not initially presumed nonnegative. Composite expressions in these displays are represented by the existing quadratic circuit.

For integers α, β, γ, put

$$\mathcal{G}(\alpha, \beta, \gamma) = \alpha^2 + \beta^2 + \gamma(\gamma + 1).$$

This is always nonnegative and represents every nonnegative integer. Indeed, the three-square theorem applies to $4n + 1$; in a representation by three squares exactly one coordinate is odd. Writing it as $2\gamma + 1$ and the other two as $2\alpha, 2\beta$ gives the assertion. The predecessor uses three separate such representations for r, s, o.

The new source deletes these nine private coordinates and imposes only

$$\boxed{rs = \mathcal{G}(\alpha, \beta, \gamma), \qquad q = (6\zeta + 3)^2.} \tag{6}$$

There are four fresh unrestricted integer coordinates here. We use the weak product condition $rs \geq 0$; an unnecessary strict condition would change the polynomial. Both equations are quadratic in the actual source registers.

2.1 Power meanings are established before the missing signs

The retained source has eighteen genuine Pell power requests, including

$$V = (F^2 + 1)^t, \quad N = B^x, \quad q = (BH_0)^x, \quad D_t = q^t, \quad D_{t+1} = q^{t+1}, \quad D_y = q^{y+1}, \quad M = q^x,$$

the factorial numerator $U_f = (q + 1)^{F^2+t}$, and the representative numerator requests. These displays describe the meanings to be proved, not extra operations in the polynomial.

The signed frame, the squared semantic cap, and the canonical orientation argument are inherited from the ordinary proofs `PROOF_315.md`, `ORIENTATION_FIRST_PROOF.md`, and `STRICT_BUDGET_PROOF.md`, retained under `source/inherited_common/upstream/round17/direct/`. Before any arithmetic sign is used, they give

$$W \geq 258, \qquad |\text{every power atom}| < W, \qquad |A| > W^W, \qquad |A - 1| > 4W^3.$$

They bound the absolute classified Pell indices, including at negative centers and negative nominal indices.

We retain all fourteen orientation terms. For a principal pair (X_i, Y_i) with requested index k_i, the pool has the form

$$\sum_{i=1}^{14} ((Y_i - k_i)^2 + 1 - X_i) = (A - 1)Q. \tag{7}$$

The actual summands $1 - X_i$ need not be nonnegative. The proof reduces them modulo $A - 1$: the small abscissa residues are $s_i \in \{-1, 1\}$, and the small ordinate residues are signed classified indices b_i. Thus the integer

$$E = \sum_{i=1}^{14} (b_i - k_i)^2 + \sum_{i=1}^{14} (1 - s_i)$$

is nonnegative, divisible by $A - 1$, and satisfies $E < 4W^3 < |A - 1|$. It is zero. This recovers the full canonical pairs. The smaller selective nine-term pool from a previous packet is *not* substituted: its later sign bootstrap would not justify the new order of argument.

2.2 The inherited deterministic comparison center

We combine (6) with the already proved deterministic-center construction [3]. The common Pell pair satisfies

$$H^2 - (A^2 - 1)J^2 = 1, \qquad J = C^2(2e + 1).$$

Replacing the old free comparison-center constructor by $f = (1 - A)H$ gives

$$D = A + Hf = A + (1 - A)H^2 = 1 - (A - 1)(A^2 - 1)J^2.$$

Consequently $D \equiv A \pmod{H}$ and $D \equiv 1 \pmod{C}$. The comparison ordinate congruence can be $R = h + Cq_c$, without the old factor 4. This deletes one additional private coordinate. It is valid for signed D, which is often negative.

The exact proof, retained as `proofs/PELL_CENTER_PRIOR.md`, uses squared-ordinate divisibility and abscissa residue separation. At an actual anchor index $i \geq 2$, Pell growth gives $i + |h| < |C|$; the congruence $i \equiv \pm h \pmod{|C|}$ therefore yields $i = |h|$. At the exceptional unit ordinate $|C| = 1$, the squared cap forces $h = 0$ and all other actual indices to vanish; the anchor then contributes a positive term to the bounded orientation residue E, a contradiction. The proof works with the full fourteen-term pool used here. Completeness chooses an even semantic anchor and rebuilds the private comparison witnesses.

2.3 The normalized digit equations

The retained common-remainder block contains the literal ratio rows

$$D_t v_t = M, \qquad D_{t+1} v_{t+1} = M, \qquad D_y v_y = M, \tag{8}$$

and the factorial numerator row

$$v_t U_f = M(B + qk_f) + R_f. \tag{9}$$

The product qk_f is already a circuit register. All eleven normalized remainders satisfy the one norm identity

$$M^2 = 1 + 4 \sum_{i=1}^{11} R_i^2 + \mathcal{G}(\alpha_*, \beta_*, \gamma_*). \tag{10}$$

The starred coordinates are independent of the new coordinates in (6). Equation (10) implies $|R_i| < M/2$ once $M > 0$. The retained normalized-remainder proof supplies stronger $1/32$ margins for completeness.

3 The new step: recover the digit, then its signs

Lemma 2 (Canonical powers at signed bases)**.** *Once the retained full orientation theorem has been applied, a power request at a base n with $|n| \geq 2$ forces its requested index k to be nonnegative and its output to be n^k.*

Proof. The ordinary power graph uses the modulus $T = 2An - n^2 - 1$. The frame gives $|T| > W^W + W$, and $T \equiv -1 \pmod{n}$, so n is invertible modulo T. For $k \geq 0$, the canonical Pell recurrence yields $m \equiv n^k \pmod{T}$. The bound on all atoms and indices makes the absolute difference smaller than $|T|$, hence $m = n^k$. For $k = -j < 0$, the inverse recurrence instead gives $mn^j \equiv 1 \pmod{T}$. The same bound would imply $mn^j = 1$, impossible for $|n| \geq 2$ and $j \geq 1$. The argument uses absolute values throughout, so negative n causes no change. At $n = 1$ and a known nonnegative index the output is 1; at $n = -1$ it is $(-1)^k$. We will exclude base zero before using the self-radix request. □

Lemma 3 (An early exponent bound)**.** *Every integer zero of the new source satisfies*

$$q \geq 9, \quad M = q^x > 0, \quad x \geq t + 1 \geq 17, \quad V \geq 1, \quad y \geq 0, \quad a_i \geq 0, \quad H_0 \geq 1, \quad B \neq 0.$$

Proof. The square in (6) is the square of an integer congruent to 3 modulo 6, hence $q \geq 9$. The genuine q^x request and Lemma 2 imply $x \geq 0$ and $M = q^x$. Since $t \geq 16$, the $t + 1$ request is also exact. The middle row of (8) is therefore

$$q^{t+1} v_{t+1} = q^x.$$

If $x < t + 1$, cancellation would make a multiple of q equal to 1. Thus $x \geq t + 1$. This uses no factorial identity or ceiling sign.

The known nonnegative t and the base $F^2 + 1 \geq 1$ give $V = (F^2 + 1)^t \geq 1$, hence $y = V(V - 1) \geq 0$. Every actual representative is an exponent in a genuine request at $q + 1 \geq 10$, so each is nonnegative. In particular $C_0 \geq 0$ and $H_0 \geq 1$. Finally $B\mu = F^2 + C_0 + 1 > 0$ excludes $B = 0$ without assigning a sign to μ. □

Lemma 4 (The self-radix bounds a signed output)**.** *At every integer zero,*

$$q = (BH_0)^x = |BH_0|^x \geq 2^x, \qquad B^2 \leq q, \qquad |B| < q/2.$$

Proof. The base BH_0 is nonzero. If its absolute value were 1, its canonical output at the already nonnegative index x would be 1 or -1, contradicting $q \geq 9$. Thus its absolute value is at least 2 and Lemma 2 applies. Positivity of q gives the equality with $|BH_0|^x$. Since $x \geq 17$ and $H_0 \geq 1$, we have $B^2 \leq q$. Finally $\sqrt{q} < q/2$ for $q \geq 9$. □

Lemma 5 (A uniform coefficient bound)**.** *Put $m = F^2$ and $n = m + t$. Then*

$$n \leq x - 1, \qquad 2^n \leq q/2.$$

Proof. The coefficient $1 + (t + 1)y$ in (4) is positive. Since $x \geq 17$, its integer quotient u_t is at least 1, giving

$$x \geq 2 + (t + 1)y. \tag{11}$$

If $m = 0$, then $n = t \leq x - 1$ by Lemma 3. If $m \geq 1$, Bernoulli's inequality yields

$$V = (m + 1)^t \geq 1 + mt \geq m + t = n.$$

Here $V \geq 2$, so $y = V(V - 1) \geq V \geq n$, and (11) is stronger than the required bound. Now $q \geq 2^x$ proves $2^n \leq 2^{x-1} \leq q/2$. □

Proposition 6 (Digit-first sign recovery)**.** *Every integer zero of the new source satisfies*

$$B = \binom{F^2 + t}{t} > 0, \qquad r \geq 0, \qquad s \geq 0, \qquad F \geq 1, \qquad B \geq 2, \qquad o \geq 0.$$

Proof. The factorial numerator is exactly $U_f = (q+1)^n$. Let

$$\rho = \sum_{j=0}^{t-1} \binom{n}{j} q^j, \qquad d = \binom{n}{t}.$$

The sum of all binomial coefficients is 2^n. Since the omitted coefficients include a positive one, Lemma 5 gives

$$0 \le \rho \le (2^n - 1)q^{t-1} < q^t/2,$$
$$0 < d < 2^n \le q/2.$$

In particular ρ is the ordinary low remainder of U_f modulo q^t. Since $v_t = M/q^t > 0$, the canonical normalized low remainder $v_t\rho$ lies in $[0, M/2)$. Equation (9) gives $R_f \equiv v_t\rho \pmod{M}$, whereas (10) gives $|R_f| < M/2$. Their difference is a multiple of M of absolute value less than M, so $R_f = v_t\rho$.

After subtraction and division by M, the digit row implies $B \equiv d \pmod{q}$. Lemma 4 gives $|B| < q/2$, and $0 < d < q/2$. Hence $|B - d| < q$ and the congruence is equality. This establishes the positive digit before the product guard has been used.

Now $rs = \mathcal{G}(\alpha, \beta, \gamma) \ge 0$ and $r + s = B - 1 \ge 0$ imply $r, s \ge 0$: opposite strict signs contradict the product, two negative signs contradict the sum, and a negative sign paired with zero also contradicts the sum. The ceiling row gives $BF = V + s > 0$, so $F \ge 1$. Moreover $V = (F^2+1)^t > F^2$, whence $BF > F^2$ and $B > F \ge 1$. Finally $q = (BH_0)^x \ge B^x > B$, so $o = q - B - 1 \ge 0$. □

All three deleted scalar guards can now be filled by fresh ternary representations. The order of the proof is essential: it does not assume the ceiling signs in order to obtain the binomial digit that establishes those signs. The retained ratio for q^{t+1}, previously serving the common-remainder normalization, supplies the early information that breaks this dependence.

4 Returning to the calibrated source

For completeness of the mathematical exposition, we give the factorial fixed-point step explicitly. The rest of the CRT and Horn-flow decoding is retained without alteration.

Lemma 7 (The square-output factorial fixed point)**.** *For the positive values recovered above, $F = t!$.*

Proof. Write $T = t!$. From $r = B - s - 1 \ge 0$ and $s \ge 0$ we have $0 \le s \le B - 1$, hence

$$B(F-1) < V \le BF, \qquad F = \lceil V/B \rceil .$$

The recovered binomial identity gives

$$\frac{V}{B} = T \prod_{i=1}^{t} \frac{F^2+1}{F^2+i} \le T.$$

As T is integral, $F \le T$. Each factor in the product is nondecreasing as a function of $F^2 \ge 1$. Evaluating at $F^2 = 1$ therefore gives

$$\frac{V}{B} \ge \frac{2^t}{t+1} > t^2 \qquad (t \ge 16).$$

The final inequality holds at 16 and is preserved upon increasing t: $2t^2 > (t+1)(t+2)$ for $t \ge 4$. Thus $F > t^2$.

For $0 \le e_i < 1$, the elementary product inequality $\prod(1-e_i) \ge 1 - \sum e_i$ now gives

$$\prod_{i=1}^{t} \frac{F^2+1}{F^2+i} = \prod_{i=1}^{t} \left(1 - \frac{i-1}{F^2+i}\right) > 1 - \frac{t(t-1)}{2F^2} > 1 - \frac{1}{2F}.$$

Consequently $T(1-1/(2F)) < V/B \le F$, so

$$T < \frac{2F^2}{2F-1} = F + \frac{F}{2F-1} < F+1.$$

Since F, T are integers, $T \le F$. Together with $F \le T$ this proves equality. □

For $1 \le i \le t$, put $d_i = 1 + F^2/i$. These are integers and $B = \prod_i d_i$. Every prime factor of d_i exceeds t: if $p \le t$, then $p \mid F^2/i$, since $F = t!$, so $p \nmid d_i$. A common prime divisor of d_i, d_j would divide $id_i - jd_j = i - j$, impossible when $i \ne j$. Thus the d_i are pairwise coprime. These are the inherited amplified CRT moduli.

After the front signs and factorial have been recovered, the proof in `source/inherited_common/proofs/NORMALIZED_REMAINDERS_PROOF.md` applies to the remaining ten low remainders. It covers the zero representative case using $\binom{a}{j} \le \max(1,a)^k$, rather than an invalid bound by 0^k. For each relevant denominator $L_i \mid M$, its genuine remainder r_i^0 satisfies $0 \le r_i^0 < L_i/32$. The norm and congruence then force $R_i = (M/L_i)r_i^0$ exactly. This restores all old quotient and interval meanings.

The unchanged prime-power range tests and unit-weight Horn-flow decoding give a finite C222 derivation of the contradiction code. The retained calibration supplies the two links

$$\mathsf{C222} \text{ derives contradiction} \iff \mathsf{ZF} \text{ has a finite refutation} \iff \mathsf{ZFC} \text{ has a finite refutation.}$$

Their named endpoints are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The corresponding Lean source files and the inherited ordinary calibration documents are included under `source/inherited_common/upstream/round17/architecture/proofs/inherited/calibration/`. This proves soundness of the new source.

5 Completeness of the square restriction

It is not enough to observe that (6) is sound. The new restriction $q = (6\zeta+3)^2$ must still allow a witness for every finite refutation.

Begin with a finite refutation and the predecessor's cofinal arithmetic construction. Choose a permitted horizon $t = 16 + h_0^2$ large enough, put $F = t!$, $V = (F^2+1)^t$, and $y = V(V-1)$. The beta-CRT moduli $1 + (i+1)y$ are all odd. A CRT residue class modulo their odd least common multiple therefore contains even, arbitrarily large integers. Choose x even and sufficiently large to satisfy all the retained endpoint bounds.

Since $i \mid F$ and $6 \mid F$, each $F^2/i = F(F/i)$ is divisible by 6. Thus

$$B = \prod_{i=1}^{t}(1 + F^2/i) \equiv 1 \pmod 6, \qquad N = B^x \equiv 1 \pmod 6, \qquad N^2 \equiv 1 \pmod 6. \tag{12}$$

The independent representative moduli have product N^2. Adding a multiple of N^2 to any actual representative preserves all its encoded local data and permits an arbitrary residue modulo 6.

One must preserve the semantic anchor while making this adjustment. Choose the ordinary b representative first. If a_0 is the least nonnegative residue for the dominant a representative, take

$$a = a_0 + 8N^2 \quad \text{or} \quad a = a_0 + 9N^2$$

to make a, b have opposite parities. The literal derived-label constructor has $h \geq 2a^2$ and $h \equiv 1 + a + b \pmod 2$. Hence h is even and at least $128N^4$.

Reserve one ordinary representative other than a, b. Start it in $(0, N^2]$, then add jN^2 for an appropriate $j \in \{0, \ldots, 5\}$ to arrange

$$H_0 = 1 + \sum_i a_i \equiv 3 \pmod 6. \tag{13}$$

The adjusted representative remains at most $6N^2$; the dominant a remains below $10N^2$. The other representatives can remain below $2N^2$, or $6N^2$ if harmless earlier parity choices are retained. The fixed front indices are below N^2, as in the parent. Thus the even $h \geq 128N^4$ still dominates every other requested index, and the squared-cap completeness estimates remain valid. The actual radix and the private frame are built after these lifts; there is no attempt to keep an obsolete upper bound on H_0.

By (12) and (13), BH_0 is an odd multiple of 3. As x is positive and even, $Q = (BH_0)^{x/2}$ is also an odd multiple of 3. Therefore

$$\zeta = (Q - 3)/6 \in \mathbb{Z}, \qquad q = (BH_0)^x = (6\zeta + 3)^2.$$

No fixed 3-adic valuation is being imposed; every odd multiple of 3 is congruent to 3 modulo 6.

The canonical ceiling remainders r, s are nonnegative, so their product has a $\mathcal{G}$ representation. The eleven genuine normalized remainders satisfy $0 \leq R_i < M/32$. Hence

$$4\sum_{i=1}^{11} R_i^2 < \frac{11}{256} M^2 < M^2.$$

The integer $M^2 - 1 - 4\sum R_i^2$ is nonnegative and has its own independent $\mathcal{G}$ representation, supplying (10). Finally choose the frame and the canonical private Pell witnesses, including the inherited deterministic comparison center. All new source rows are satisfied.

This is an existential construction with fresh private witnesses. It does not claim a bijection of all old and new solution tuples. Together with Section 4, it proves both directions of the source equivalence.

6 From the source theorem to the literal quartic

The source compiler and certificates are included so that the displayed equation is a finite object with a reviewable connection to the proof. No search for witnesses or assumption of consistency is used to construct its coefficients.

6.1 Exact algebraic transformations

The file `source/equation247_weak/delta_certificate.json` records every changed source row and every deleted private guard. The source audit checks the actual polynomials, not just row names. Eighteen inherited Horn factorizations and 204 integral monic restorations lead to the raw 247-variable system; seventy integral translations give its centered coordinates.

A monic restoration deletes a variable only when an explicit polynomial with integer coefficients reconstructs it from the retained variables. The reverse substitution is checked exactly. Thus there are no unreported divisibility conditions or rational coordinates. Unused private coordinates are omitted only after their nonoccurrence has been verified.

The final coordinate census explains the net saving precisely. The nine old scalar-guard coordinates and the old free center coordinate disappear. Four fresh guard coordinates appear, and the coordinate `fact0.remainder`, which was eliminable in the predecessor, is retained to keep the new product row quadratic. Thus the measured change is $252 - 10 + 5 = 247$, rather than a naive count of deleted guards.

To improve the final coefficient profile, we transfer 76 known integral coordinate shears from the prior shared-residue packet [5], matching the original coordinate names rather than assuming the same coordinate numbering. A step (i, j, k) means

$$\text{old } z_j = \text{new } z_j + k \,\text{new } z_i.$$

It has determinant one and inverse coefficient $-k$. The entire integer matrix, its two-sided inverse, and the full old-to-new coordinate map are checked. These are coordinate changes on $\mathbb{Z}^{247}$, so integer solvability is preserved.

Four additional residual operations improve the height:

$$R_{18} \leftarrow R_{18} - 22R_{11}, \quad R_{19} \leftarrow R_{19} - R_{11}, \quad R_{52} \leftarrow R_{52} - R_{11}, \quad R_{27} \leftarrow R_{27} - R_{23}.$$

The indices here are one-based and the operations occur in the displayed order. They are recorded with zero-based indices in `certificates/row_operations.json`. Each operation is an invertible integer row operation. It preserves the simultaneous zero set of the residuals. It generally changes the scalar sum-of-squares polynomial, so we do not assert an identity between the old and new sums of squares.

All final residuals remain quadratic, with maximum absolute residual coefficient 66 and 1,077 total residual monomial occurrences. Their squared sum has the exact metrics in Section 1; all 247 variables occur in the collected polynomial.

6.2 What has been checked

`source/verify_composed.py` independently reconstructs the source delta, invokes the retained independently implemented Horn and monic replay, checks translations, and collects the raw quartic. `code/build_final.py` reconstructs the final finite outputs from the frozen coordinate and residual certificates.

`code/check_final.py` is a second implementation using sparse exponent-vector monomials and explicit binomial expansion. It imports no producer or eliminator. It checks each map forward and backward, both matrix inverse identities, every final residual, every collected coefficient, the complete coordinate map, degree, height, support, and coordinate liveness. It also parses the actual LaTeX residual file back into polynomials and compares them with the machine-readable equation.

Six deliberate source/projection corruptions and eight deliberate final-algebra corruptions are rejected. The latter include changes to a shear, its historical coordinate mapping, a residual operation, a final residual, a collected coefficient, the coordinate map, a printed formula, and the domain. Normal Python and optimized Python both pass; disabling assertions does not disable the checks. The receipts and substantive reviews are retained in `review/`.

These finite checks prove exact algebraic identities for the emitted finite data. They do not replace the unbounded number-theoretic arguments or recheck the entire inherited Lean development. The proof of Theorem 1 is the ordinary argument in Sections 2–5, composed with the retained calibration and the verified finite transformations. Every new argument uses elementary arithmetic, finite coding and standard set-theoretic constructions formalizable in ZFC. No fresh kernel-checked proof of this complete final theorem, and no encoded object-ZFC proof term, is claimed.

7 Reproducibility, attribution, and further research

The canonical starting point is `README.md`. With Python's standard library, run

```
python3 -B verify.py
python3 -O -B verify.py
```

The default checks are read-only. To reconstruct the final files from the frozen raw source and maps, run

```
python3 -B code/build_final.py --write
```

To evaluate an explicitly supplied 247-integer tuple without pretending to decide consistency, run

```
python3 -B code/evaluate.py tuple.json --expanded
```

The input is a JSON list in the order $z_1, \ldots, z_{247}$. The build script `build_report.sh` compiles this report and the optional experimental note. The package includes the ordinary source proofs, exact transformation certificates, independent checkers, machine-readable equations, all report inputs, a provenance ledger, and a SHA-256 manifest.

7.1 The actual new contribution

The deterministic Pell center, odd-square idea, common normalized remainder norm, and historical coordinate shears are credited to their incoming source packets [3, 4, 2, 5]. The new mathematical contribution here is the specific *digit-first sign recovery* argument:

$$q^{t+1} \mid q^x \implies x \geq t+1 \implies 2^{F^2+t} \leq q/2 \implies B = \binom{F^2+t}{t} > 0.$$

The self-radix bound on $|B|$ is also essential in the last implication. This order permits the weak product guard and removes the three individual front sign certificates. The nonunit square $(6\zeta+3)^2$ and its cofinal CRT construction make the early radix argument unconditional.

This is a concrete improvement relative to the audited snapshot. It is not an exhaustive literature-priority claim or a declaration of a major breakthrough. General universal-equation results [6, 7, 8] establish useful context for variable-count tradeoffs but do not supply this direct source theorem.

7.2 A different calculus was implemented and measured

The accompanying directory `experiments/three_parameter_calculus/` investigates replacing the four-parameter implicational schema by K, S, and Peirce, together with a modular saturation equation whose inverse coordinate is unbounded. A literal 521-step Hilbert derivation certifies the propositional replacement. Its note proves the required local small-value decoding, prime-divisor argument, CRT inverse choice, and nonnegative quotient repair.

The resulting experimental quartic has 231 *nonnegative* integer variables, 183 residuals, 6,369 monomials and height 83,899,166. It loses to its own 229-variable nonnegative parent: extra source branches and multiplication gates cost more than the removed bounded column saves. These domains and architectures must not be confused with the 247 unrestricted integers of Theorem 1. The experiment is included because it provides an exact, falsifiable result for a distinct proposed route. Its standalone checker verifies the finite basis certificate and polynomial expansion; rebuilding through its different full parent compiler additionally requires the pinned parent archive identified in that directory.

7.3 Questions suggested by the reduction

The most promising next work is to examine existing divisibility identities for information available before the sign bootstrap. The q^{t+1} ratio is indispensable to the present proof; deleting it on the assumption that it is merely a remainder-normalization overhead would destroy the early exponent bound. Other ratios may have a different status, but a reduction must preserve both quadratic degree and an integral restoration.

One can also ask whether the single product nonnegativity condition can be represented with fewer coordinates under additional cofinal arithmetic restrictions. Such a claim needs a complete cofinal construction; a sign argument alone is insufficient. The unsuccessful calculus experiment suggests

co-designing the local proof calculus with its arithmetic circuit, since fewer logical parameters need not mean fewer final polynomial variables.

Finally, the integer coordinate and residual searches can be improved independently of the semantics, provided every chosen map has a checked integral inverse. Formalizing the early-exponent and digit-sign lemmas in the existing proof environment would make a valuable next integration step. None of these proposed directions is used as an unproved premise in the delivered equation.

References

[1] V. Reshetnikov et al., *ProveIt: Computability and Diophantine consistency research*, snapshot `3ecd6a6cf53d47519228328567f45fb7ee9781d9`. https://github.com/VladimirReshetnikov/ProveIt/tree/3ecd6a6cf53d47519228328567f45fb7ee9781d9/Computability/DiophantineConsistency/con-zfc-diophantine-forms. The ledger also pins the incoming archives.

[2] *ConZFC Common Remainders / Radix 47*, 25 October 2026, `docs/incoming/ConZFC_Common_Remainders_Radix47_2026-10-25.zip`, at snapshot [1]. The portable direct source closure and its ordinary proofs are retained under `source/inherited_common/`.

[3] *ConZFC Symmetric Remainders*, 25 October 2026, `docs/incoming/ConZFC_Symmetric_Remainders_2026-10-25.zip`, at snapshot [1]. Deterministic comparison-center proof retained as `proofs/PELL_CENTER_PRIOR.md`.

[4] *ConZFC Symmetric Residue Budgets*, 25 October 2026, `docs/incoming/ConZFC_Symmetric_Residue_Budgets_2026-10-25.zip`, at snapshot [1]. Prior odd-square proof retained as `proofs/ODD_SQUARE_PRIOR.md`.

[5] *ConZFC Shared Residue Bounds*, 25 October 2026, `docs/incoming/ConZFC_Shared_Residue_Bounds_2026-10-25.zip`, at snapshot [1]. Exact historical shear inputs and transfer certificate are retained in `certificates/`.

[6] J. P. Jones, Universal Diophantine equation, *Journal of Symbolic Logic* 47 (1982), 549–571. https://doi.org/10.2307/2273588.

[7] J. Bayer, M. David, M. Hassler, Y. Matiyasevich and D. Schleicher, *Diophantine Equations over $\mathbb{Z}$: Universal Bounds and Parallel Formalization*, arXiv:2506.20909, version 3, 22 October 2026. https://arxiv.org/abs/2506.20909.

[8] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594, version 2, 2 September 2026. https://arxiv.org/abs/2609.01594.

A All 182 quadratic residuals

Every symbol below is one of the 247 unrestricted integer coordinates. The single equation is the sum of the squares of these complete residuals, set equal to zero. Coefficients omitted before a monomial are 1 or -1 according to the displayed sign. There are no additional definitions or constraints in this appendix.

$$
\begin{aligned}
R_{1} &= -z_{4}^{2} + z_{5} \\
R_{2} &= -1 - z_{4}\,z_{6} + z_{6} + z_{7} \\
&\quad - z_{8} \\
R_{3} &= z_{1} - z_{1}\,z_{3} - z_{3} + z_{11} \\
R_{4} &= -1 + z_{2} - z_{3}\,z_{9} - z_{4} \\
R_{5} &= -1 + z_{2} - z_{10}\,z_{11} \\
R_{6} &= 1 + z_{5} - z_{6}\,z_{22} + z_{20} \\
R_{7} &= -1 + z_{12} - z_{12}^{2} - 2\,z_{12}\,z_{13} \\
&\quad - z_{13} - z_{13}^{2} + z_{23} \\
R_{8} &= -5 - 20\,z_{12} - 16\,z_{12}^{2} - 16\,z_{12}\,z_{13} \\
&\quad - 16\,z_{12}\,z_{23} - 10\,z_{13} - 4\,z_{13}^{2} - 8\,z_{13}\,z_{23} \\
&\quad - 10\,z_{23} - 4\,z_{23}^{2} + z_{24} \\
R_{9} &= -2 - z_{12} - z_{12}^{2} - 4\,z_{12}\,z_{24} \\
&\quad - 4\,z_{24} - 4\,z_{24}^{2} + z_{25} \\
R_{10} &= 1 + 2\,z_{25} - 4\,z_{25}^{2} + z_{26} \\
R_{11} &= -2 - z_{13}^{2} - 2\,z_{13}\,z_{14} - z_{14} \\
&\quad - z_{14}^{2} + z_{27} \\
R_{12} &= -9 + 7\,z_{12} - z_{12}^{2} - 4\,z_{12}\,z_{13} \\
&\quad + 4\,z_{12}\,z_{14} - 4\,z_{12}\,z_{27} + 12\,z_{13} - 4\,z_{13}^{2} \\
&\quad + 8\,z_{13}\,z_{14} - 8\,z_{13}\,z_{27} - 15\,z_{14} - 4\,z_{14}^{2} \\
&\quad + 8\,z_{14}\,z_{27} + 12\,z_{27} - 4\,z_{27}^{2} + z_{28} \\
&\quad - z_{34} \\
R_{13} &= -1 + z_{12} - z_{12}^{2} - 2\,z_{12}\,z_{14} \\
&\quad - z_{13} - z_{14}^{2} + z_{29} \\
R_{14} &= -4\,z_{12} - 64\,z_{12}^{2} - 64\,z_{12}\,z_{14} - 32\,z_{12}\,z_{23} \\
&\quad - 32\,z_{12}\,z_{29} - 2\,z_{13} - 16\,z_{14}^{2} - 16\,z_{14}\,z_{23} \\
&\quad - 16\,z_{14}\,z_{29} - 2\,z_{23} - 4\,z_{23}^{2} - 8\,z_{23}\,z_{29} \\
&\quad - z_{28} - 4\,z_{29}^{2} + z_{30} + z_{34} \\
R_{15} &= -36\,z_{14}^{2} + 24\,z_{14}\,z_{30} - 4\,z_{30}^{2} + z_{31}
\end{aligned}
$$

$$
\begin{aligned}
R_{16} = &-6 - 8\,z_{12} - 4\,z_{12}^2 - 8\,z_{12}\,z_{13} \\
&- 8\,z_{12}\,z_{14} + 8\,z_{12}\,z_{27} + 8\,z_{12}\,z_{29} - 6\,z_{13} \\
&- 4\,z_{13}^2 - 8\,z_{13}\,z_{14} + 8\,z_{13}\,z_{27} + 8\,z_{13}\,z_{29} \\
&- 5\,z_{14} - 4\,z_{14}^2 + 8\,z_{14}\,z_{27} + 8\,z_{14}\,z_{29} \\
&- z_{23} + 8\,z_{27} - 4\,z_{27}^2 - 8\,z_{27}\,z_{29} \\
&+ 6\,z_{29} - 4\,z_{29}^2 + z_{32}
\end{aligned}
$$

$$
\begin{aligned}
R_{17} = &\,1 - 4\,z_{12} + 2\,z_{13} - 4\,z_{13}^2 \\
&- 8\,z_{13}\,z_{14} - 8\,z_{13}\,z_{32} + 4\,z_{14} - 4\,z_{14}^2 \\
&- 8\,z_{14}\,z_{32} - 2\,z_{23} + 4\,z_{32} - 4\,z_{32}^2 \\
&+ z_{33}
\end{aligned}
$$

$$
\begin{aligned}
R_{18} = &\,34 - 5\,z_{12} - z_{12}^2 - 12\,z_{12}\,z_{13} \\
&- 12\,z_{12}\,z_{14} - 4\,z_{12}\,z_{27} - 36\,z_{13} - 14\,z_{13}^2 \\
&- 28\,z_{13}\,z_{14} - 24\,z_{13}\,z_{27} - 11\,z_{14} - 14\,z_{14}^2 \\
&- 24\,z_{14}\,z_{27} - 34\,z_{27} - 4\,z_{27}^2 + z_{34}
\end{aligned}
$$

$$
\begin{aligned}
R_{19} = &-16\,z_{12} - 16\,z_{12}^2 + 8\,z_{12}\,z_{13} - 32\,z_{12}\,z_{14} \\
&- 16\,z_{12}\,z_{29} + 5\,z_{13} + 10\,z_{13}\,z_{14} + 4\,z_{13}\,z_{29} \\
&- 14\,z_{14} - 15\,z_{14}^2 - 16\,z_{14}\,z_{29} - z_{27} \\
&- 8\,z_{29} - 4\,z_{29}^2 - z_{34} + z_{35}
\end{aligned}
$$

$$
\begin{aligned}
R_{20} = &\,2 - 64\,z_{14}^2 - 32\,z_{14}\,z_{35} - 4\,z_{35}^2 \\
&+ z_{36}
\end{aligned}
$$

$$
\begin{aligned}
R_{21} = &-12 - 3\,z_{12} - 47\,z_{13} - 36\,z_{13}^2 \\
&+ 24\,z_{13}\,z_{23} + 24\,z_{13}\,z_{27} - z_{14} + 16\,z_{23} \\
&- 4\,z_{23}^2 - 8\,z_{23}\,z_{27} + 14\,z_{27} - 4\,z_{27}^2 \\
&- z_{29} + z_{37}
\end{aligned}
$$

$$
\begin{aligned}
R_{22} = &-3 - 6\,z_{12} + 2\,z_{13} - 2\,z_{14} \\
&- 2\,z_{29} - 4\,z_{37} - 4\,z_{37}^2 + z_{38}
\end{aligned}
$$

$$
\begin{aligned}
R_{23} = &-1 - 13\,z_{12} - 16\,z_{12}^2 - 16\,z_{12}\,z_{13} \\
&- 16\,z_{12}\,z_{23} - 6\,z_{13} - 4\,z_{13}^2 - 8\,z_{13}\,z_{23} \\
&- 6\,z_{23} - 4\,z_{23}^2 + z_{39}
\end{aligned}
$$

$$
\begin{aligned}
R_{24} = &-2 + 3\,z_{12} - z_{12}^2 + 4\,z_{12}\,z_{39} \\
&- 4\,z_{39} - 4\,z_{39}^2 + z_{40}
\end{aligned}
$$

$$
R_{25} = 1 + 2\,z_{40} - 4\,z_{40}^2 + z_{41}
$$

$$
R_{26} = -z_{13}^2 - 4\,z_{13}\,z_{41} - 4\,z_{41}^2 + z_{42}
$$

$$
\begin{aligned}
R_{27} = &-5 - 3\,z_{12} + 48\,z_{12}\,z_{13} - 16\,z_{12}\,z_{23} \\
&+ 12\,z_{13} - 12\,z_{13}^2 + 40\,z_{13}\,z_{23} - 2\,z_{14} \\
&- 8\,z_{23} - 12\,z_{23}^2 - z_{27} - z_{39} \\
&+ z_{43}
\end{aligned}
$$

$$
\begin{aligned}
R_{28} = &-4 - 3\,z_{12} - 7\,z_{13} - 4\,z_{14} \\
&- 2\,z_{27} - z_{29} + 4\,z_{43} - 4\,z_{43}^2 \\
&+ z_{44}
\end{aligned}
$$

$$
\begin{aligned}
R_{29} &= -6\,z_{12} + 2\,z_{13} + 2\,z_{14} - 4\,z_{14}^2 \\
&\quad - 8\,z_{14}\,z_{44} - 2\,z_{29} + 4\,z_{44} - 4\,z_{44}^2 \\
&\quad + z_{45}
\end{aligned}
$$

$$
\begin{aligned}
R_{30} &= -z_{14} - z_{14}^2 - 2\,z_{14}\,z_{15} - z_{15}^2 \\
&\quad + z_{46}
\end{aligned}
$$

$$
\begin{aligned}
R_{31} &= 2\,z_{12} - z_{12}^2 - 4\,z_{12}\,z_{46} + 2\,z_{46} \\
&\quad - 4\,z_{46}^2 + z_{47} - z_{48}
\end{aligned}
$$

$$
\begin{aligned}
R_{32} &= -35 + 2\,z_{13} - 66\,z_{14} - 36\,z_{14}^2 \\
&\quad + 24\,z_{14}\,z_{27} + 24\,z_{14}\,z_{29} + 24\,z_{27} - 4\,z_{27}^2 \\
&\quad - 8\,z_{27}\,z_{29} + 22\,z_{29} - 4\,z_{29}^2 + z_{48}
\end{aligned}
$$

$$
\begin{aligned}
R_{33} &= -16\,z_{14}^2 - 16\,z_{14}\,z_{47} - 2\,z_{47} - 4\,z_{47}^2 \\
&\quad + 2\,z_{48} + z_{49}
\end{aligned}
$$

$$
\begin{aligned}
R_{34} &= -16\,z_{12}^2 + 24\,z_{12}\,z_{14} - 16\,z_{12}\,z_{23} + 16\,z_{12}\,z_{32} \\
&\quad + z_{14} - 9\,z_{14}^2 + 12\,z_{14}\,z_{23} - 12\,z_{14}\,z_{32} \\
&\quad - 4\,z_{23}^2 + 8\,z_{23}\,z_{32} - 4\,z_{32}^2 + z_{50}
\end{aligned}
$$

$$
\begin{aligned}
R_{35} &= -1 + 6\,z_{13} - 4\,z_{13}^2 + 8\,z_{13}\,z_{14} \\
&\quad - 8\,z_{13}\,z_{27} - 6\,z_{14} - 4\,z_{14}^2 + 8\,z_{14}\,z_{27} \\
&\quad + 6\,z_{27} - 4\,z_{27}^2 + z_{51}
\end{aligned}
$$

$$
\begin{aligned}
R_{36} &= -1 - z_{14} - z_{14}^2 - 4\,z_{14}\,z_{51} \\
&\quad - 4\,z_{51}^2 + z_{52}
\end{aligned}
$$

$$
\begin{aligned}
R_{37} &= 1 + z_{13} - z_{13}^2 - 4\,z_{13}\,z_{52} \\
&\quad - 4\,z_{52}^2 + z_{53}
\end{aligned}
$$

$$
R_{38} = -1 - 6\,z_{53} - 4\,z_{53}^2 + z_{54}
$$

$$
\begin{aligned}
R_{39} &= -1 - z_{14} - z_{14}^2 - 4\,z_{14}\,z_{54} \\
&\quad - 4\,z_{54}^2 + z_{55}
\end{aligned}
$$

$$
\begin{aligned}
R_{40} &= 2 + z_{13} - z_{13}^2 - 4\,z_{13}\,z_{55} \\
&\quad - 4\,z_{55}^2 + z_{56}
\end{aligned}
$$

$$
\begin{aligned}
R_{41} &= -2\,z_{50} - 4\,z_{50}^2 - 8\,z_{50}\,z_{56} - 4\,z_{56}^2 \\
&\quad + z_{57}
\end{aligned}
$$

$$
\begin{aligned}
R_{42} &= -9 + 7\,z_{12} - z_{12}^2 - 4\,z_{12}\,z_{57} \\
&\quad + 12\,z_{57} - 4\,z_{57}^2 + z_{58}
\end{aligned}
$$

$$
R_{43} = -1 - 6\,z_{58} - 4\,z_{58}^2 + z_{59}
$$

$$
\begin{aligned}
R_{44} &= -z_{14} - z_{14}^2 - 4\,z_{14}\,z_{59} - 4\,z_{59}^2 \\
&\quad + z_{60}
\end{aligned}
$$

$$
\begin{aligned}
R_{45} &= -z_{12} - z_{12}^2 - 4\,z_{12}\,z_{60} - 4\,z_{60} \\
&\quad - 4\,z_{60}^2 + z_{61}
\end{aligned}
$$

$$
\begin{aligned}
R_{46} &= -10 - z_{13} + z_{14} - z_{27} \\
&\quad - 14\,z_{61} - 4\,z_{61}^2 + z_{62}
\end{aligned}
$$

$$R_{47} = 1 - 2\,z_{13} + 2\,z_{14} - 2\,z_{27} + 4\,z_{62} - 4\,z_{62}^2 + z_{63}$$

$$R_{48} = z_{13} - z_{13}^2 - 4\,z_{13}\,z_{63} - 4\,z_{63}^2 + z_{64} - z_{65}$$

$$R_{49} = -10 - 14\,z_{56} - 4\,z_{56}^2 + z_{65}$$

$$R_{50} = 1 - 2\,z_{64} - 4\,z_{64}^2 + 2\,z_{65} + z_{66}$$

$$R_{51} = z_{14} - z_{14}^2 - 4\,z_{14}\,z_{66} - 4\,z_{66}^2 + z_{67}$$

$$R_{52} = 2 + 15\,z_{13} - 35\,z_{13}^2 + 14\,z_{13}\,z_{14} + 24\,z_{13}\,z_{23} - 4\,z_{14}\,z_{23} - 5\,z_{23} - 4\,z_{23}^2 - z_{27} + z_{68}$$

$$R_{53} = 1 + 6\,z_{13} - 2\,z_{23} - z_{67} + 4\,z_{68} - 4\,z_{68}^2 + z_{69}$$

$$R_{54} = -2 + 2\,z_{67} - 2\,z_{69} - 4\,z_{69}^2 + z_{70}$$

$$R_{55} = -2 - 4\,z_{70} - 4\,z_{70}^2 + z_{71}$$

$$R_{56} = -z_4\,z_{17} + z_{72}$$

$$R_{57} = z_{16} - z_{17} - 2\,z_{26}\,z_{72} - z_{73}$$

$$R_{58} = 12\,z_{14}\,z_{72} + z_{16} - z_{17} - 4\,z_{28}\,z_{72} - 2\,z_{31}\,z_{72} + 4\,z_{34}\,z_{72} - z_{74}$$

$$R_{59} = z_{16} - z_{17} - 2\,z_{33}\,z_{72} - 6\,z_{72} - z_{75}$$

$$R_{60} = -12\,z_{14}\,z_{72} + z_{16} - z_{17} - 4\,z_{34}\,z_{72} - 2\,z_{36}\,z_{72} - z_{76}$$

$$R_{61} = z_{16} - z_{17} - 2\,z_{38}\,z_{72} + 6\,z_{72} - z_{77}$$

$$R_{62} = z_{16} - z_{17} - 4\,z_{41}\,z_{72} - 2\,z_{42}\,z_{72} - z_{78}$$

$$R_{63} = z_{16} - z_{17} - 2\,z_{45}\,z_{72} - z_{79}$$

$$R_{64} = z_{16} - z_{17} - 2\,z_{49}\,z_{72} - 4\,z_{72} - z_{80}$$

$$R_{65} = -z_{12}\,z_{13} + z_{12}\,z_{14} + z_{13} + z_{13}\,z_{14} - z_{14} - z_{14}^2 + z_{81}$$

$$R_{66} = 1 + 2\,z_{15} + z_{15}^2 - z_{81}^2 + z_{82}$$

$$R_{67} = -z_{82}^2 + z_{83}$$

$$R_{68} = -4\,z_{70}\,z_{72} - 2\,z_{71}\,z_{72} + 2\,z_{72} + z_{84}$$

$$R_{69} = -z_{16}^2 + 2\,z_{16}\,z_{17} + 2\,z_{16}\,z_{84} - z_{17}^2 - 2\,z_{17}\,z_{84} - z_{84}^2 + z_{85}$$

$$R_{70} = -1 - z_4\,z_{13} + z_{86}$$

$$R_{71} = -z_{17}\,z_{86} + z_{88}\,z_{89} + z_{90}$$

$$R_{72} = -1 + z_4 - z_4\,z_{12} + z_{87}$$

$$R_{73} = -1 + 2\,z_4 + 6\,z_4\,z_{13} - 2\,z_4\,z_{23} + z_{88}$$

$$R_{74} = -z_{16}\,z_{87} + z_{89}$$

$$R_{75} = -1 - 4\,z_4\,z_{12} - 2\,z_4\,z_{13} - 2\,z_4\,z_{23} + z_{91}$$

$$R_{76} = z_{16}\,z_{86} - z_{17}\,z_{91} + z_{92}$$

$$R_{77} = -z_{16}\,z_{73} + z_{17}\,z_{73} + z_{93}$$

$$R_{78} = -z_{74}\,z_{93} + z_{94}$$

$$R_{79} = -z_{75}\,z_{94} + z_{95}$$

$$R_{80} = -z_{76}\,z_{95} + z_{96}$$

$$R_{81} = -z_{77}\,z_{96} + z_{97}$$

$$R_{82} = -z_{78}\,z_{97} + z_{98}$$

$$R_{83} = -z_{79}\,z_{98} + z_{99}$$

$$R_{84} = -z_{80}\,z_{99} + z_{100}$$

$$R_{85} = -z_{90}\,z_{100} + z_{101}$$

$$R_{86} = -z_{92}\,z_{101} + z_{102}$$

$$R_{87} = -z_{102}^2 + z_{103}$$

$$R_{88} = -z_3 - z_3\,z_{20} + z_{20} + z_{104}$$

$$R_{89} = -z_{18}\,z_{104} + z_{105}$$

$$R_{90} = -2\,z_2^2 + 2\,z_2\,z_{16} + 2\,z_2\,z_{17} + 2\,z_2\,z_{105} + 2\,z_2\,z_{106} - z_{16}^2 - 2\,z_{16}\,z_{105} - z_{17}^2 - 2\,z_{17}\,z_{106} + z_{21}\,z_{108} - z_{83}\,z_{103} - z_{85}\,z_{103} - z_{105}^2 - z_{106}^2$$

$$R_{91} = -z_3\,z_{19} + z_{19} - z_{19}\,z_{104} + z_{106}$$

$$R_{92} = -5\,z_{12} + 3\,z_{13} - 4\,z_{23} - 2\,z_{26} - 4\,z_{28} - 2\,z_{31} - 2\,z_{33} - 2\,z_{36} - 2\,z_{38} - 4\,z_{41} - 2\,z_{42} - 2\,z_{45} - 2\,z_{49} - 4\,z_{70} - 2\,z_{71} + z_{107}$$

$$R_{93} = -z_{109}\,z_{112} + z_{112} + z_{114}$$

$$R_{94} = -z_6\,z_{136} + z_{110}\,z_{235} - z_{113} - z_{114}\,z_{136}$$

$$R_{95} = z_{115}\,z_{236} - z_{117}\,z_{139} - z_{118}$$

$$R_{96} = z_{119}\,z_{236} - z_{120}\,z_{139} - z_{121}$$

$$R_{97} = z_{122}\,z_{236} - z_{123}\,z_{139} - z_{124}$$

$$R_{98} = z_{125}\, z_{236} - z_{126}\, z_{139} - z_{127}$$

$$R_{99} = z_{128}\, z_{237} - z_{130}\, z_{139} - z_{131}$$

$$R_{100} = z_{132}\, z_{237} - z_{133}\, z_{139} - z_{134}$$

$$R_{101} = -z_{21}\, z_{136} + z_{139}$$

$$R_{102} = z_{135} - z_{137}\, z_{139} - z_{138}$$

$$R_{103} = -z_{139}\, z_{141} + z_{140} - z_{142}$$

$$R_{104} = -z_{139}\, z_{144} + z_{143}\, z_{235} - z_{145}$$

$$R_{105} = -z_{139}\, z_{147} + z_{146}\, z_{236} - z_{148}$$

$$R_{106} = z_{149} - 2\, z_{149}\, z_{151} + z_{152}$$

$$R_{107} = -z_{149}\, z_{152} - z_{152} + z_{153}$$

$$R_{108} = 2\, z_{150} + z_{150}^2 + z_{152}^2 - z_{153}^2$$

$$R_{109} = -z_{150}\, z_{155} - z_{155} + z_{156}$$

$$R_{110} = 2\, z_{154} + z_{154}^2 + z_{155}^2 - z_{156}^2$$

$$R_{111} = z_5^2 - 2\, z_5\, z_{150} - 2\, z_{150} + z_{157}$$

$$R_{112} = -1 - z_5\, z_{155} + z_7 - z_{154} - z_{155} + z_{156} + z_{157}\, z_{158}$$

$$R_{113} = 1 - 2\, z_6 + z_6^2 - 2\, z_6\, z_{150} + z_{159}$$

$$R_{114} = -z_6\, z_{205} + z_{21} + z_{159}\, z_{160} - z_{204} + z_{206}$$

$$R_{115} = -1 + 2\, z_{150}\, z_{220} - z_{161} + 2\, z_{220} - z_{220}^2$$

$$R_{116} = -1 + z_{109} + z_{161}\, z_{162} - z_{204} - z_{205}\, z_{220} + z_{206}$$

$$R_{117} = -z_{150}\, z_{164} - z_{164} + z_{165}$$

$$R_{118} = 2\, z_{163} + z_{163}^2 + z_{164}^2 - z_{165}^2$$

$$R_{119} = 1 - 2\, z_{109} + z_{109}^2 - 2\, z_{109}\, z_{150} + z_{166}$$

$$R_{120} = -1 - z_{109}\, z_{164} + z_{110} - z_{163} + z_{165} + z_{166}\, z_{167}$$

$$R_{121} = 4 - 4\, z_{109} + z_{109}^2 - 2\, z_{109}\, z_{150} + 2\, z_{150} + z_{168}$$

$$R_{122} = -1 - z_{109}\, z_{155} + z_{111} - z_{154} + z_{155} + z_{156} + z_{168}\, z_{169}$$

$$R_{123} = 1 + z_{150} - z_{150}\, z_{171} - z_{171} + z_{172}$$

$$R_{124} = 1 + 2\, z_{170} + z_{170}^2 - 2\, z_{171} + z_{171}^2 - z_{172}^2$$

$$R_{125} = -1 + z_{109} - z_{109}\,z_{171} + z_{115} + z_{166}\,z_{173} - z_{170} + z_{172}$$

$$R_{126} = -2\,z_{150} - z_{150}^2 + z_{174}$$

$$R_{127} = 1 - z_{109} - z_{109}\,z_{154} - z_{109}\,z_{156} + z_{116} + z_{150}\,z_{156} + z_{154} - z_{155}\,z_{174} + 2\,z_{156} + z_{168}\,z_{175}$$

$$R_{128} = -z_{150}\,z_{177} - z_{177} + z_{178}$$

$$R_{129} = 2\,z_{176} + z_{176}^2 + z_{177}^2 - z_{178}^2$$

$$R_{130} = -1 - z_{109}\,z_{177} + z_{119} + z_{166}\,z_{179} - z_{176} + z_{178}$$

$$R_{131} = -z_{150}\,z_{181} - z_{181} + z_{182}$$

$$R_{132} = 2\,z_{180} + z_{180}^2 + z_{181}^2 - z_{182}^2$$

$$R_{133} = -1 - z_{109}\,z_{181} + z_{122} + z_{166}\,z_{183} - z_{180} + z_{182}$$

$$R_{134} = -z_{150}\,z_{185} - z_{185} + z_{186}$$

$$R_{135} = 2\,z_{184} + z_{184}^2 + z_{185}^2 - z_{186}^2$$

$$R_{136} = -1 - z_{109}\,z_{185} + z_{125} + z_{166}\,z_{187} - z_{184} + z_{186}$$

$$R_{137} = -z_{150}\,z_{189} - z_{189} + z_{190}$$

$$R_{138} = 2\,z_{188} + z_{188}^2 + z_{189}^2 - z_{190}^2$$

$$R_{139} = -1 - z_{109}\,z_{189} + z_{128} + z_{166}\,z_{191} - z_{188} + z_{190}$$

$$R_{140} = -z_{150}\,z_{193} - z_{193} + z_{194}$$

$$R_{141} = 2\,z_{192} + z_{192}^2 + z_{193}^2 - z_{194}^2$$

$$R_{142} = -1 - z_{109}\,z_{193} + z_{129} + z_{168}\,z_{195} - z_{192} + z_{193} + z_{194}$$

$$R_{143} = -z_{150}\,z_{197} - z_{197} + z_{198}$$

$$R_{144} = 2\,z_{196} + z_{196}^2 + z_{197}^2 - z_{198}^2$$

$$R_{145} = -1 - z_{109}\,z_{197} + z_{132} + z_{166}\,z_{199} - z_{196} + z_{198}$$

$$R_{146} = -z_{150}\,z_{201} - z_{201} + z_{202}$$

$$R_{147} = 2\,z_{200} + z_{200}^2 + z_{201}^2 - z_{202}^2$$

$$R_{148} = -1 - z_{109}\,z_{201} + z_{135} + z_{166}\,z_{203} - z_{200} + z_{202}$$

$$R_{149} = 1 - z_{150}\,z_{205} - z_{205} + z_{206}$$

$$R_{150} = -1 + 2\,z_{204} + z_{204}^2 + z_{205}^2 - 2\,z_{206} - z_{206}^2$$

$$R_{151} = -z_{109}\, z_{205} + z_{136} + z_{168}\, z_{207} - z_{204} + z_{205} + z_{206}$$

$$R_{152} = -z_{150}\, z_{209} - z_{209} + z_{210}$$

$$R_{153} = 2\, z_{208} + z_{208}^2 + z_{209}^2 - z_{210}^2$$

$$R_{154} = -1 - z_{109}\, z_{209} + z_{140} + z_{166}\, z_{211} - z_{208} + z_{210}$$

$$R_{155} = -z_{150}\, z_{213} - z_{213} + z_{214}$$

$$R_{156} = 2\, z_{212} + z_{212}^2 + z_{213}^2 - z_{214}^2$$

$$R_{157} = -1 - z_{109}\, z_{213} + z_{143} + z_{166}\, z_{215} - z_{212} + z_{214}$$

$$R_{158} = -z_{150}\, z_{217} - z_{217} + z_{218}$$

$$R_{159} = 2\, z_{216} + z_{216}^2 + z_{217}^2 - z_{218}^2$$

$$R_{160} = -1 - z_{109}\, z_{217} + z_{146} + z_{166}\, z_{219} - z_{216} + z_{218}$$

$$\begin{aligned} R_{161} = {} & -6 - 2\, z_1 - 3\, z_1^2 - 2\, z_1\, z_5 \\ & - z_2^2 - z_3^2 - 2\, z_5 - 2\, z_5^2 \\ & - z_6^2 - z_7^2 + 2\, z_{12} - z_{12}^2 \\ & - z_{13}^2 - z_{14}^2 - z_{15}^2 - z_{16}^2 \\ & - z_{17}^2 - z_{18}^2 - z_{19}^2 - z_{20}^2 \\ & - z_{21}^2 - z_{107}^2 + 2\, z_{109} - 2\, z_{109}^2 \\ & - z_{110}^2 - z_{111}^2 - z_{115}^2 - z_{116}^2 \\ & - z_{119}^2 - z_{122}^2 - z_{125}^2 - z_{128}^2 \\ & - z_{129}^2 - z_{132}^2 - z_{135}^2 - z_{136}^2 \\ & - z_{140}^2 - z_{143}^2 - z_{146}^2 + z_{149} \\ & - z_{220}^2 \end{aligned}$$

$$R_{162} = -z_6\, z_{12} - z_6\, z_{13} - z_6\, z_{14} - z_6\, z_{15} - z_6\, z_{16} - z_6\, z_{17} - z_6\, z_{18} - z_6\, z_{19} - z_6\, z_{20} + z_{220}$$

$$R_{163} = -16 + z_1 - z_{221}^2$$

$$R_{164} = -1 + z_3 + z_7 - z_7^2$$

$$R_{165} = -z_{150}\, z_{223} - z_{223} + z_{231}$$

$$R_{166} = -1 + z_{222}^2 + z_{223}^2 - z_{231}^2$$

$$R_{167} = -z_{225} - z_{225}\, z_{226} + z_{232}$$

$$R_{168} = -1 + z_{224}^2 + z_{225}^2 - z_{232}^2$$

$$R_{169} = -z_{217}^2 + z_{233}$$

$$R_{170} = z_{223} - 2\, z_{227}\, z_{233} - z_{233}$$

$$R_{171} = z_{150}\, z_{222} + z_{228}$$

$$R_{172} = -z_{150} - z_{222}\, z_{228} + z_{226}$$

$$R_{173} = -1 - z_{216} - z_{222}\, z_{229} + z_{224}$$

$$R_{174} = -z_{107} - z_{217}\, z_{230} + z_{225}$$

$$\begin{aligned} R_{175} = {} & 2\, z_1^2 + 2\, z_1\, z_5 - 2\, z_1\, z_{155} - 2\, z_1\, z_{164} \\ & + z_2^2 - 2\, z_2\, z_{205} + z_3^2 - 2\, z_3\, z_{193} \\ & + z_5^2 - 2\, z_5\, z_{164} + z_{12}^2 - 2\, z_{12}\, z_{171} \\ & + z_{13}^2 - 2\, z_{13}\, z_{177} + z_{14}^2 - 2\, z_{14}\, z_{181} \\ & + z_{15}^2 - 2\, z_{15}\, z_{185} + z_{16}^2 - 2\, z_{16}\, z_{189} \\ & + z_{17}^2 - 2\, z_{17}\, z_{197} + z_{18}^2 - 2\, z_{18}\, z_{201} \\ & + z_{19}^2 - 2\, z_{19}\, z_{209} + z_{20}^2 - 2\, z_{20}\, z_{213} \\ & + z_{107}^2 - 2\, z_{107}\, z_{217} - z_{150}\, z_{234} - z_{154} \\ & + z_{155}^2 - z_{163} + z_{164}^2 - z_{170} \\ & + z_{171}^2 - z_{176} + z_{177}^2 - z_{180} \\ & + z_{181}^2 - z_{184} + z_{185}^2 - z_{188} \\ & + z_{189}^2 - z_{192} + z_{193}^2 - z_{196} \\ & + z_{197}^2 - z_{200} + z_{201}^2 - z_{204} \\ & + z_{205}^2 - z_{208} + z_{209}^2 - z_{212} \\ & + z_{213}^2 - z_{216} + z_{217}^2 \end{aligned}$$

$$R_{176} = -z_{111}\, z_{235} + z_{136}$$

$$R_{177} = -z_{116}\, z_{236} + z_{136}$$

$$R_{178} = -z_{129}\, z_{237} + z_{136}$$

$$\begin{aligned} R_{179} = {} & -2 - z_{107}^2 - z_{155}^2 - z_{164}^2 \\ & + 2\, z_{171} - z_{171}^2 - z_{177}^2 - z_{181}^2 \\ & - z_{185}^2 - z_{189}^2 - z_{193}^2 - z_{197}^2 \\ & - z_{201}^2 - z_{205}^2 - z_{209}^2 - z_{213}^2 \\ & + z_{217}^2 - z_{238}^2 - z_{239}^2 - z_{240} \\ & - z_{240}^2 \end{aligned}$$

$$\begin{aligned} R_{180} = {} & -1 - 4\, z_{113}^2 - 4\, z_{118}^2 - 4\, z_{121}^2 \\ & - 4\, z_{124}^2 - 4\, z_{127}^2 - 4\, z_{131}^2 - 4\, z_{134}^2 \\ & + z_{136}^2 - 4\, z_{138}^2 - 4\, z_{142}^2 - 4\, z_{145}^2 \\ & - 4\, z_{148}^2 - z_{241}^2 - z_{242}^2 - z_{243} \\ & - z_{243}^2 \end{aligned}$$

$$R_{181} = -10 + z_{109} - 36\, z_{244} - 36\, z_{244}^2$$

$$\begin{aligned} R_{182} = {} & z_6\, z_8 - z_8 - z_8^2 - z_{245}^2 \\ & - z_{246}^2 - z_{247} - z_{247}^2 \end{aligned}$$

B Exact data format and integrity

In a polynomial JSON file a term is a pair $[c, [i_1, \ldots, i_d]]$, denoting $c\, z_{i_1+1} \cdots z_{i_d+1}$. Indices are zero-based in the files and one-based in the displayed mathematics. An empty index list is a constant, repeated indices are powers, all coefficients are integers, and all like monomials have been collected in the expansion.

The SHA-256 digest of the final residual equation file is

`6df8f7de8b389084d6ce000f777f27ffc962e05b51217662e680201cffd1578d`

and that of the collected polynomial is

`19b2bc4e8cbb8d112feed0b88ca6c91884a40985745026eed020a78eee35187f.`

The period after the second digest is punctuation. The raw 247-variable weak-product equation before the final integral maps has digest

`14b814894521add09fb6847e59de2d802f2e466f6f9fc724f346a21461125a6b.`

The hashes bind finite files; they do not constitute a mathematical proof. The manifest separately binds all deliverable files, excluding the manifest itself.